Experiment 23

Aim

To perform **image sharpening using the Unsharp Masking technique**, which enhances the edges and fine details of an image by subtracting a blurred version of the original image.

Procedure

1. Read the input image.
2. Convert the image to grayscale if required.
3. Apply Gaussian blur to the original image to obtain a blurred image.
4. Subtract the blurred image from the original image to obtain the unsharp mask.
5. Add the unsharp mask to the original image.
6. Display the original, blurred, unsharp mask, and sharpened images.
7. Save the sharpened image.

The basic operation is:

Unsharp Mask = Original Image − Blurred Image

Sharpened Image = Original Image + Unsharp Mask

Program

```
import cv2
import matplotlib.pyplot as plt

# Read input image
img = cv2.imread("input.jpg")

# Convert BGR to RGB for displaying
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Apply Gaussian Blur
blurred = cv2.GaussianBlur(img, (5, 5), 0)

# Create unsharp mask
```

```
unsharp_mask = cv2.subtract(img, blurred)

# Sharpen the image
sharpened = cv2.add(img, unsharp_mask)

# Convert images to RGB
blurred_rgb = cv2.cvtColor(blurred, cv2.COLOR_BGR2RGB)
mask_rgb = cv2.cvtColor(unsharp_mask, cv2.COLOR_BGR2RGB)
sharpened_rgb = cv2.cvtColor(sharpened, cv2.COLOR_BGR2RGB)

# Display results
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(blurred_rgb)
plt.title("Blurred Image")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(mask_rgb)
plt.title("Unsharp Mask")
plt.axis("off")

plt.subplot(2, 2, 4)
```

```
plt.imshow(sharpened_rgb)
plt.title("Sharpened Image")
plt.axis("off")

plt.tight_layout()
plt.show()

# Save sharpened image
cv2.imwrite("sharpened_image.jpg", sharpened)
```

Input

Output